AquaVoice

One-page solution pitch | Great Indian AI Internship Challenge | November 2026

Problem

Aquaculture farmers need timely, understandable guidance about water quality, weather and market conditions. Digital advisory tools can be difficult to use when information is technical or language is a barrier. AquaVoice explores a regional-language, voice-first way to access aquaculture decision support.

Solution

AquaVoice is a voice-first extension of AquaGuard AI, an existing Python/Streamlit multi-agent aquaculture decision-support application. It is designed for farmers to submit short voice notes and receive context-aware guidance in a preferred Indian language. The prototype combines a farm-risk dashboard, an audio-upload workflow, a manual transcript fallback for testing the reasoning path, and a Telugu response experience.

How it works

1. The farmer views the available farm profile and condition information in AquaGuard. 2. AquaVoice accepts a short audio file and a language fallback selection. 3. The intended speech-to-text step uses Gnani AI Prisma v3.0. 4. A manual transcript test can independently pass text to the AquaGuard/Gemini reasoning workflow. 5. A Telugu text-to-speech step is integrated using Gnani AI Timbre v3.0.

Technology and model integrations

Application: Python, Streamlit and the existing AquaGuard multi-agent workflow. **Reasoning:** Gemini is used as the supporting reasoning model for the manual-transcript handoff. **Gnani integrations:** Prisma v3.0 (speech-to-text) and Timbre v3.0 (text-to-speech) are implemented in the application using the documented API routes.

Current test status and transparency

A manually entered Telugu transcript was successfully passed to the AquaGuard/Gemini workflow, which returned a Telugu answer. This is a manual-transcript/Gemini test, not a Gnani transcription. During live testing, requests to both Gnani Prisma and Timbre endpoints were rejected with HTTP 403 / Cloudflare Error 1010. Therefore, the application currently demonstrates the integrations and UI, but successful live inference from either Gnani endpoint has not been verified. The issue has been reported to Gnani AI support. No live IoT sensor connection is claimed; stored farm records are treated as historical context rather than current telemetry.

Expected impact and next step

The intended impact is to make aquaculture guidance more accessible through regional-language voice interaction while preserving transparency about data freshness and model limitations. The immediate next step is to verify live Prisma and/or Timbre access with Gnani AI support and complete an end-to-end audio test.

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